Rank-refined enumeration of upper triangular symplectic Renner monoids

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Editorial note. One rank-refined enumeration object mainTheorem. Complete original mathematical document90–1515, all admissible-set/rook-matrix bijections, recurrence/generating/Laguerre proofs, original tables and embedded bibliography retained. Native original end-document is1516; post-document source text1517–1655 remains in the raw archived author file and is not inserted into the rendered document or counted.

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Allowed standard prerequisites (editorial).

- Injective partial transformations, symplectic rook matrices, admissible sets and rank enumeration. Original monoid, admissibility, identity exception, boundary conventions and rank/dimension parameter ranges retained exactly. Classical algebraic-monoid background and Laguerre generating facts remain original cited prerequisites.. Complete original admissible-set bijections, rank recurrence and subsequent generating/closed-form proofs supplied; no independent matrix enumeration or proof reconstruction.

1 Introduction

The set R_m of all injective partial transformations of $\mathbf{m} = \{1, 2, \ldots, m\}$ is a monoid (semigroup with an identity) under composition of maps, with identity element $\mathrm{id}_{\mathbf{m}}$. An

injective partial transformation of m is a one-to-one map of a subset (domain) of m onto a subset (range) of m.

Let M_m^* denote the set of all $m \times m$ matrices $A = [a_{ij}]$ such that each $a_{ij} = 0$ or 1 for all i, j, and such that each row and each column contains at most one 1. Then, M_m^* is a monoid under matrix multiplication with identity I_m. There is a monoid isomorphism between R_m and M_m^*, whereby $a_{ij} = 1$ iff $f(j) = i$. We identify R_m with M_m^* throughout.

For instance, let $\mathbf{m} = \{1, 2, 3, 4\}$. The matrix

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in M_4^*$$

corresponds to the injective partial transformation $f \in \mathrm{R}_4$ with domain $D(f) = \{1, 3, 4\}$ given by the column indices of the nonzero entries, and range $R(f) = \{1, 2, 4\}$ given by the row indices of the nonzero entries. In map notation

$$1 \mapsto 2, \qquad 2 \notin D(f), \qquad 3 \mapsto 1, \qquad 4 \mapsto 4.$$

In combinatorics, these matrices can be viewed as placements of any number of non-attacking rooks on an $m \times m$ chessboard, and R_m is consequently called the rook monoid. We use the terms injective partial transformation and rook matrix interchangeably, according to the context. The submonoids B_m of R_m consisting of upper triangular injective partial transformations of $\mathbf{m}$ are called upper triangular rook monoids.

The orders

$$|\,\mathrm{R}_m\,| = \sum_{r=0}^{m} \binom{m}{r}^2 r!$$

are closely related to the Laguerre polynomials, which have been described as one of the "prettiest gems" of combinatorial theory [1, 2]. The orders of B_m are the Stirling numbers of the second kind, which arise naturally in the study of set partitions. The exponential generating functions of both $|\,\mathrm{R}_m\,|$ and $|B_m|$ are given in [2].

It is sometimes useful to regard R_m, where $m \geqslant 1$, as the Renner monoid of the general linear algebraic monoid $M_m(K)$ in the theory of linear algebraic monoids [6, 7, 8, 9], where K is an algebraically closed field.

The symplectic Renner monoids RSp_n, where $n = m/2$ and m is even, form another infinite family of rook matrices of size $2n \times 2n$, consisting of symplectic injective partial transformations (Definition 2.6). A systematic study of their structures and cardinalities using the symplectic Weyl group and admissible subsets (Definition 2.1) can be found in [3, 5].

The orders $|\,\mathrm{RSp}_n\,|$ and the exponential generating functions of the symplectic Renner monoids were studied in [4]. In particular,

$$|\,\mathrm{RSp}_n\,| = \sum_{r=0}^{n} 4^r \binom{n}{r}^2 r! + 2^n n!.$$

These orders are also closely connected to the Laguerre polynomials [4].

Let $\mathbb{B}_n$ denote the subset of RSp_n consisting of upper triangular symplectic injective partial transformations. These form submonoids of RSp_n, called the upper triangular symplectic Renner monoids. For $0 \leq r \leq 2n$, define

$$\mathbb{B}_{n,\,r} = \{f \in \mathbb{B}_n \;\big|\; |D(f)| = r\}, \quad 0 \leqslant r \leqslant 2n.$$

Thus, $\mathbb{B}_{n,r}$ consists of the elements of $\mathbb{B}_n$ whose domains have cardinality r.

We investigate the orders and exponential generating functions of $\mathbb{B}_n$ and $\mathbb{B}_{n,r}$. In particular, we derive recursive formulas for calculating $\mathbb{B}_n$ and $\mathbb{B}_{n,r}$ in Theorem 3.5. These formulas are then used to obtain the exponential generating functions for b_n and $b_{n,\,n-r}$ in Theorems 4.1 and 4.5, respectively. We further establish connections between these generating functions and the Laguerre polynomials in Corollary 4.3. We then provide a recurrence relation of order three for the sequence b_n in Theorem 4.4. Closed formulas for b_n and $b_{n,r}$ are given in Theorems 4.2 and 4.6, respectively.

The remainder of the paper is organized as follows. Section 2 introduces the concepts and techniques involving admissible sets and upper triangular symplectic Renner monoids. Section 3 develops recursive formulas for the orders of these monoids and gives numerical values for the first seven cases. Section 4 presents the exponential generating functions and closed formulas for b_n and $b_{n,r}$. It also provides a recurrence relation for the sequence b_n involving the preceding three terms, b_{n-1}, b_{n-2}, and b_{n-3}. Finally, Section 5 presents a conjecture that may be useful for further study of the generating functions of b_n.

2 Preliminaries

Let $\mathbf{m} = \{1, 2, ..., m\}$ with $m \geqslant 1$. Define an involution θ of $\mathbf{m}$ by

$$\theta(i) = m + 1 - i, \quad i \in \mathbf{m}.$$

Then $\theta \in \mathrm{R}_m$. The involution θ in $\mathbf{m} = \mathbf{2n}$ can be interpreted geometrically as reflection across the 'median line', illustrated here with $n = 4$:

1	2	3	4	5	6	7	8

Definition 2.1. A subset I of $\mathbf{m}$ is said to be admissible if $I = \mathbf{m}$ or if $\theta(I) \cap I = \varnothing$. An admissible subset not equal to $\mathbf{m}$ is called proper admissible.

If $m = 2n$ with $n \geqslant 1$, we use $\mathcal{A}_n$ to denote the set of all admissible subsets of $\mathbf{2n}$. Note that the empty set $\varnothing$ is admissible. If $\mathbf{m} = \{1, 2\}$, then

$$\mathcal{A}_1 = \{\varnothing,\ \{1\},\ \{2\},\ \{1, 2\}\}.$$

If $\mathbf{m} = \{1, 2, 3, 4\}$, then

$$\mathcal{A}_2 = \{\varnothing,\ \{1\},\ \{2\},\ \{3\},\ \{4\},\ \{1, 2\},\ \{1, 3\},\ \{2, 4\},\ \{3, 4\},\ \{1, 2, 3, 4\}\}.$$

The two lemmas below, presented without proof, state some simple properties of admissible sets.

Lemma 2.2. Let $\mathbf{m} = \{1, 2, ..., m\}$ with $m \geqslant 1$. Then the intersection of two admissible sets is admissible, and each subset of a proper admissible set is admissible.

Lemma 2.3. Let m be even and $i, j \in \mathbf{m} = \{1, 2, \ldots, m\}$. We have

a) $\theta(i) \neq i$ and hence the set $\{i\}$ is admissible.

b) If $j \neq \theta(i)$, then $\{i, j\}$ is admissible.

c) The set $\{i, \theta(i)\}$ is not admissible.

d) If I is a proper admissible set, then $|I| \leqslant m/2$.

The set $\mathcal{A}_n^{\times} = \mathcal{A}_n \setminus \{\mathbf{2n}\}$ consisting of all proper admissible subsets of $\mathbf{2n}$ is useful. The following lemma will be used in Section 3.

Lemma 2.4. Let $n \geqslant 2$, let $\tau\colon \{2, \ldots, 2n-1\} \to \{1, \ldots, 2n-2\}$ be the map defined by $\tau(k) = k - 1$, and let $\mathcal{B}_n = \{I \in \mathcal{A}_n^{\times} \mid I \cap \{1, 2n\} = \varnothing\}$.

a) If $I \subseteq \{2, \ldots, 2n-1\}$, then $I \in \mathcal{B}_n$ if and only if $\tau(I) \in \mathcal{A}_{n-1}^{\times}$.

b) If $K \in \mathcal{A}_{n-1}^{\times}$ then both $\tau^{-1}(K) \cup \{1\}$ and $\tau^{-1}(K) \cup \{2n\}$ are in $\mathcal{A}_n^{\times}$.

Proof. We prove a). Let θ be the involution of $\mathbf{2n}$, and let θ' be the involution of $\mathbf{2n-2}$, so that $\theta(i) = 2n+1-i$ $(1 \leqslant i \leqslant 2n)$, and $\theta'(k) = 2n-1-k$ $(1 \leqslant k \leqslant 2n-2)$. It is easy to verify that $\theta(i) - 1 = \theta'(i-1)$ for $2 \leqslant i \leqslant 2n-1$. Let I be a non-empty set in $\mathcal{B}_n$, and let $J = \tau(I)$. Suppose to the contrary that J is not admissible. Then there exist i, $j \in I$ such that $\theta'(i-1) = j-1$. It follows that $\theta(i) = j$ contradicting the fact that I is admissible. The proof of the converse is similar.

We now prove b). If $K \in \mathcal{A}_{n-1}^{\times}$ then from a) it follows that $\tau^{-1}(K) \subseteq \{2, \ldots, 2n-1\}$ is in $\mathcal{A}_n^{\times}$. Since $\theta(1)$ and $\theta(2n)$ are both not in $\tau^{-1}(K)$ it follows that $\tau^{-1}(K) \cup \{1\}$ and $\tau^{-1}(K) \cup \{2n\}$ are in $\mathcal{A}_n^{\times}$. □

Recall that R_m is the monoid of all injective partial transformations of $\mathbf{m} = \{1, 2, ..., m\}$. The domain and range of $f \in \mathrm{R}_m$ are denoted by $D(f)$ and $R(f)$, respectively.

Definition 2.5. Let f be an element of R_m. If $D(f)$ is the entire set $\mathbf{m}$, then f is called a full transformation of $\mathbf{m}$. If $D(f)$ is a proper subset of $\mathbf{m}$, then f is called a proper injective partial transformation of $\mathbf{m}$.

2.1 Symplectic Renner Monoids

To investigate the theory of symplectic Renner monoids, we require that $m = 2n$, with $n \geqslant 1$.

Definition 2.6. An injective partial transformation f in R_{2n} is said to be symplectic if $D(f)$ is admissible and f takes admissible subsets of $D(f)$ to admissible subsets of $\mathbf{2n}$. The set of all symplectic transformations is denoted by

$$\mathrm{RSp}_n = \{f \in \mathrm{R}_{2n} \mid f \text{ is symplectic}\}.$$

The identity element $\mathrm{id}_{\mathbf{2n}}$ of R_{2n} and the involution θ lie in RSp_n. Note that a full transformation in R_{2n} is symplectic if it takes admissible subsets of $\mathbf{2n}$ to admissible subsets of $\mathbf{2n}$, that is, it preserves $\mathcal{A}_n$. It is evident that $\mathrm{id}_{\mathbf{2n}}$ is full symplectic. Furthermore, θ is full symplectic. First note that θ is bijective. Next if I is proper admissible in $\mathbf{2n}$ and $J = \theta(I)$, then $J \cap \theta(J) = \theta(I) \cap I = \varnothing$. Thus, $\theta(I)$ is admissible.

The full injective transformation

$$h = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

is not symplectic, since $I = \{2, 4\}$ is admissible yet $h(I) = \{2, 3\}$ is not admissible.

Combining Lemmas 2.2 and 2.3 d), we obtain the following result, which will be used implicitly throughout the rest of the paper.

Lemma 2.7. a) A proper injective partial transformation is symplectic iff both its domain and range are admissible.

b) There are no proper symplectic injective partial transformations with domain size between $n + 1$ and $2n - 1$ inclusive.

An element f of a monoid $(M, \circ, e)$ is said to be a unit if there exists $g \in M$ such that $fg = gf = e$, the identity element of M. The set of the units of M is a group, called the unit group of M.

The group S_{2n} of all bijective transformations of $\mathbf{2n}$ is the unit group of R_{2n}. In fact, S_{2n} consists precisely of the non-singular rook matrices of size $2n \times 2n$. Both S_{2n} and R_{2n} share the same identity element $\mathrm{id}_{\mathbf{2n}}$.

Let $f \in \mathrm{R}_{2n}$. Its (partial) inverse, denoted by f^{-1}, is defined by:

$$f^{-1}(b) = a \quad \text{if} \quad f(a) = b, \quad \text{where } a \in D(f), b \in R(f).$$

If f is full, then $f \in S_{2n}$ and its partial inverse f^{-1} is the inverse of f in S_{2n}.

Lemma 2.8. a) If $f \in \mathrm{RSp}_n$, then $f^{-1} \in \mathrm{RSp}_n$.

b) If $f, g \in \mathrm{RSp}_n$, then $gf \in \mathrm{RSp}_n$.
Thus, RSp_n is a submonoid of R_{2n}.

Proof. a) Let f be symplectic. If f is full, then f preserves admissible sets, hence all powers f^k $(k \geqslant 1)$ are symplectic. Note that f is in S_{2n}; we deduce that f^{-1} is symplectic, since f^{-1} is a power of f. If f is proper, then both $D(f)$ and $R(f)$ are proper admissible. But then $D(f^{-1}) = R(f)$ and $R(f^{-1}) = D(f)$. Thus, f^{-1} is symplectic.

b) Let f, g be symplectic and $h = gf$, where f is applied first. Suppose both f and g are full. Then h is full, and since f and g preserve admissible sets, h is symplectic. Next, suppose at least one of f and g is proper. Then h is proper, and $K = R(f) \cap D(g)$ is proper admissible, whence $D(h) = f^{-1}(K)$ is proper admissible by a), and $R(h) = g(K)$ is proper admissible. Thus, $h \in \mathrm{RSp}_n$. We note that if K is empty then h is the zero matrix.

Since $\mathrm{id}_{2n} \in \mathrm{RSp}_n$, it follows that RSp_n is a submonoid of R_{2n}. □

The monoids RSp_n are called the symplectic Renner monoids in the theory of reductive algebraic monoids. See [3, 4, 5, 6] for more details.

Lemma 2.9. Let $W_n := \mathrm{RSp}_n \cap S_{2n}$. Then W_n is the unit group of RSp_n.

Proof. Write U for the unit group of RSp_n. If $f \in U$ then $f \in \mathrm{RSp}_n$ and there exists $g \in \mathrm{RSp}_n$ such that $fg = gf = \mathrm{id}_{2n}$. Hence $f \in S_{2n}$, whence $f \in RSp_n \cap S_{2n} = W_n$. On the other hand suppose that $f \in W_n$. Then $f \in \mathrm{RSp}_n$. Since also $f \in S_{2n}$, there is $g \in S_{2n}$ such that $fg = gf = \mathrm{id}_{2n}$. By Lemma 2.8 a), $g \in \mathrm{RSp}_n$. Thus $f \in U$. □

To give a more concrete characterization of W_n, we can describe its elements in terms of their action on admissible sets:

$$\begin{aligned} W_n &= \{f \in S_{2n} \mid f \text{ takes admissible sets to admissible sets}\} \\ &= \{f \in \mathrm{R}_{2n} \mid f \text{ is full symplectic}\}. \end{aligned}$$

We then have the following disjoint union.

$$\mathrm{RSp}_n = W_n \sqcup \{f \in \mathrm{R}_{2n} \mid D(f) \text{ and } R(f) \text{ are proper adimissible}\}.$$

Thus RSp_n consists of the elements in its unit group W_n and the proper symplectic injective partial transformations.

If $n = 1$, then

$$\mathrm{RSp}_1 = \left\{ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}.$$

If $n = 2$, then RSp_2 consists of fifty seven elements [5]. The following proper injective partial transformation f lies in RSp_2, whereas g doesn't lie in RSp_2:

$$f = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad g = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

This is because $D(f) = \{1, 2\}$ and $R(f) = \{1, 3\}$ are admissible, whereas $D(g) = \{2, 3\}$ is not admissible although the range $R(g) = \{1, 3\}$ is.

Lemma 2.10. The unit group W_n of RSp_n is the centralizer of θ in S_{2n}. In other words, W_n equals

$$C_{S_{2n}}(\theta) = \{f \in S_{2n} \mid \theta f = f\theta\}.$$

Proof. We note that $\theta \in W_n$ since $\theta \in RSp_n \cap S_{2n}$. Denote $C = C_{S_{2n}}(\theta)$.

We first show that $C \subseteq W_n$. Let $f \in C$. Then $f \in S_{2n}$, thus $D(f) = \mathbf{2n}$, so it suffices to show that for any proper admissible subset A of $\mathbf{2n}$ we have $f(A)$ is admissible. Suppose for a contradiction that $\theta(f(A)) \cap f(A)$ is not empty. Then there exist i, j in A such that $\theta(f(i)) = f(j)$. As $f \in C$, it follows that $f(\theta(i)) = f(j)$, whence $\theta(i) = j$ since f is injective. Since A is admissible, this is a contradiction.

We next show that $W_n \subseteq C$. Suppose for a contradiction that there is $f \in W_n$ but $f \notin C$. Then there is $i \in \mathbf{2n}$ such that $f(\theta(i)) \neq \theta(f(i))$. It follows from Lemma 2.3 b) that the set $\{f(i), f(\theta(i))\}$ is admissible. Since $f^{-1} \in W_n$, the set $f^{-1}(\{f(i), f(\theta(i)\}) = \{i, \theta(i)\}$ is admissible, which contradicts Lemma 2.3 c). □

It turns out that W_n is isomorphic to the symplectic Weyl group in Lie theory. We sometimes just say that W_n is the symplectic Weyl group. If $n = 1$, then $\theta = (12)$ and $W_1 = C_{S_2}(\theta) = \{(1), (12)\}$. The matrix form is

$$W_1 = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}.$$

If $n = 2$, then $\theta = (14)(23)$ and

$$W_2 = C_{S_4}(\theta) = \{(1), (14), (23), (14)(23), (12)(34), (1243), (1342), (13)(24)\}$$
$$= \left\{ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \right.$$
$$\left. \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \right\}.$$

2.2 Upper Triangular Symplectic Renner Monoids $\mathbb{B}_n$

Let $\mathbb{B}_n$ be the submonoid of RSp_n consisting of all the upper triangular symplectic injective partial transformations. That is,

$$\mathbb{B}_n = \{f \in \mathrm{RSp}_n \mid f \text{ is upper triangular}\}.$$

It is evident that the unit group of $\mathbb{B}_n$ is the trivial group $\{\mathrm{id}_{\mathbf{2n}}\}$.

Definition 2.11. The submonoid $\mathbb{B}_n$ of RSp_n is called the upper triangular symplectic Renner monoid.

If $n = 1$, then

$$\mathbb{B}_1 = \left\{ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

If $n = 2$, then

$$\mathbb{B}_2 = \left\{ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \right.$$
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$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$
$$\left. \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \right\}.$$

We then have the orders $|\mathbb{B}_1| = 5$ and $|\mathbb{B}_2| = 25$.

3 Orders of $\mathbb{B}_n$

Let $\mathbf{2n} = \{1, 2, ..., 2n\}$ and $n \geqslant 1$. Recall that members of $\mathbb{B}_n$ are upper triangular symplectic rook matrices of size $2n \times 2n$.

Definition 3.1. Let $n \geqslant 1$. We define

$$\mathbb{B}_{n,r} = \{f \in \mathbb{B}_n \mid |D(f)| = r\}, \quad 0 \leqslant r \leqslant 2n.$$

By Lemma 2.7 b), $\mathbb{B}_{n,r} = \varnothing$ for $n+1 \leqslant r \leqslant 2n-1$, and $\mathbb{B}_{n,2n} = \{\mathrm{id}_{2n}\}$ since the identity element $\mathrm{id}_{\mathbf{2n}}$ is the only element in $\mathbb{B}_n$ of rank $2n$. It follows immediately that

$$\mathbb{B}_n = \bigsqcup_{r=0}^{2n} \mathbb{B}_{n,r} = \Big(\bigsqcup_{r=0}^{n} \mathbb{B}_{n,r}\Big) \bigsqcup \{\mathrm{id}_{2n}\}.$$

Define $b_{n,r} = |\mathbb{B}_{n,r}|$ for $0 \leqslant r \leqslant n$. Then $b_{n,0} = 1$. We agree that $b_{0,0} = 1$ and $b_{n,r} = 0$ if $n < r$ or $r < 0$. The following lemmas are useful for calculating the order of $\mathbb{B}_n$. Write $\bar{i} = \theta(i)$. Then $\bar{1} = 2n$.

Lemma 3.2. Let $n \geqslant 1$ and $0 \leqslant r \leqslant n$. Write

$$X_{n,r} = \{f \in \mathbb{B}_{n,r} \mid 1 \notin R(f) \text{ and } \bar{1} \notin D(f)\}.$$

Then there exists a bijection of $X_{n,r}$ onto $\mathbb{B}_{n-1,r}$. Thus, $|X_{n,r}| = b_{n-1,r}$.

Proof. If $n = 1$, then $r = 0, 1$. Thus, we have

$$X_{1,0} = \{0\}, \quad |X_{1,0}| = 1; \quad b_{0,0} = 1$$
$$X_{1,1} = \varnothing, \qquad |X_{1,1}| = 0; \quad b_{0,1} = 0.$$

Thus, the lemma holds for $0 \leqslant r \leqslant n = 1$.

We now suppose that $n \geqslant 2$. Let $0 \leqslant r \leqslant n$. If $f \in X_{n,r}$, then row 1 and column $\bar{1}$ are both zero and also row $\bar{1}$ and column 1 are zero. Therefore, f takes the form

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ & * & * & \cdots & * & 0 \\ & & * & \cdots & * & 0 \\ & & & \ddots & \vdots & \vdots \\ & & & & * & 0 \\ & & & & & 0 \end{pmatrix}_{2n \times 2n} \in X_{n,r},$$

where $*$ is either 0 or 1, each row and each column of f contains at most one 1, and $D(f)$ and $R(f)$ are of cardinality r and proper admissible.

If $r = n$, then by Lemma 2.3 d) and Lemma 2.4 we have

$$X_{n,n} = \varnothing, \quad |X_{n,n}| = 0, \quad b_{n-1,n} = 0.$$

That is, the result is true for $r = n$.

Suppose now that $0 \leqslant r \leqslant n-1$, and again $n \geqslant 2$. Let f' be obtained from $f \in X_{n,r}$ by deleting row 1, row $\bar{1}$, column 1, and column $\bar{1}$. If $f = [a_{ij}] \in X_{n,r}$, then $f' = [b_{kl}]$

where $b_{k\,l} = a_{k+1,\,l+1}$. By Lemma 2.4 a) both the domain and range of f' are admissible sets in $\mathcal{A}_{n-1}$. Hence $f' \in \mathbb{B}_{n-1,\,r}$. Then we have a map $\kappa : f \mapsto f'$. The map κ is onto: given $g \in \mathbb{B}_{n-1,\,r}$, then adjoining two zero rows and two zero columns on the outside of g yields an $f \in X_{n,\,r}$ for which $f' = g$. The map κ is one-to-one: if $f, g \in X_{n,\,r}$ and $f' = g'$, then $f = g$ since f indeed agrees with g on row 1, row $\bar{1}$, column 1, and column $\bar{1}$. Hence $|X_{n,\,r}| = b_{n-1,\,r}$. □

Lemma 3.3. Let $n \geqslant 1$ and $0 \leqslant r \leqslant n$. Define

$$Z_{n,\,r} = \{f \in \mathbb{B}_{n,\,r} \mid 1 \notin R(f) \text{ and } f(\bar{1}) = j \text{ for some } 2 \leqslant j \leqslant 2n-1\}.$$

Then

$$|Z_{n,\,r}| = (2n-2r)b_{n-1,\,r-1}.$$

Proof. Take $n = 1$. We have

$$\begin{aligned} n=1,\, r=0: &\quad Z_{n,\,r} = Z_{1,\,0} = \varnothing; &&\quad (2n-2r)b_{n-1,\,r-1} = 2b_{0,\,-1} = 0 \\ n=1,\, r=1: &\quad Z_{n,\,r} = Z_{1,\,1} = \varnothing; &&\quad (2n-2r)b_{n-1,\,r-1} = 0b_{0,\,0} = 0. \end{aligned}$$

The lemma holds when $n = 1$.

We now suppose that $n \geqslant 2$. If $r = 0$, then $Z_{n,\,r} = Z_{n,\,0} = \varnothing$ whereas $(2n - 2r)b_{n-1,\,r-1} = 2nb_{n-1,\,-1} = 0$, so the result is true when $r = 0$. If $r = n$, then $Z_{n,\,r} = Z_{n,\,n} = \varnothing$, whereas $(2n-2r)b_{n-1,\,r-1} = 0$, so the result is true when $r = n$.

We may now assume that $1 \leqslant r \leqslant n-1$ so that $r-1 \geqslant 0$ and $n-1 \geqslant 1$. If $f \in Z_{n,\,r}$, then f is of the form

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & \cdots & 0 & 0 \\ & * & \cdots & * & \cdots & * & 0 \\ & & \ddots & \vdots & \cdots & \vdots & \vdots \\ & & & 0 & \cdots & 0 & 1 \\ & & & & \ddots & \vdots & \vdots \\ & & & & & * & 0 \\ & & & & & & 0 \end{pmatrix}_{2n\times 2n} \leftarrow a_{j,\,2n} = 1\ ,$$

where $2 \leqslant j \leqslant 2n-1$. Deleting the first row, first column, last row, and last column of f produces an element $f' \in \mathbb{B}_{n-1,\,r-1}$.

We shall show that for each $g \in \mathbb{B}_{n-1,\,r-1}$, there are $2n-2r$ distinct f's in $Z_{n,\,r}$ such that $f' = g$. Indeed fix $g \in \mathbb{B}_{n-1,\,r-1}$, where $0 \leqslant r-1 \leqslant n-2$. Let $J = \tau^{-1}(R(g))$, where $\tau(k) = k-1$. By Lemma 2.4 a), J is admissible in $\mathcal{A}_n$, and $|J \cup \theta(J)| = 2r-2$, where θ is the involution in $\mathbf{2n}$. Let $E(g)$ denote the complement of $J \cup \theta(J)$ in $\{2, \ldots, 2n-1\}$, so that $|E(g)| = 2n - 2r \geq 2$.

For each $j \in E(g)$ we extend g by adjoining a zero row at the top and the bottom, a zero column at the left, and a column with 1 in position j at the right; this extension is

denoted by $\hat{g}_j$. The domain $D(\hat{g}_j) = \tau^{-1}(D(g)) \cup \{2n\}$ is admissible in $\mathcal{A}_n$ by Lemma 2.4 b). The range $R(\hat{g}_j) = \tau^{-1}(R(g)) \cup \{j\}$ is also admissible since $\theta(R(\hat{g}_j)) \cap R(\hat{g}_j)$ is empty, by the construction of $E(g)$ and that $\theta(j) \neq j$. Thus $\hat{g}_j$ is in $Z_{n,r}$. Moreover $\kappa(\hat{g}_j) = g$ for all $j \in E(g)$.

Next, note that if $j \neq j'$ in $E(g)$, then $\hat{g}_j \neq \hat{g}_{j'}$. Moreover, if $g \neq h$ in $\mathbb{B}_{n-1,r-1}$, then for each $j \in E(g)$ and $k \in E(h)$ we have $\hat{g}_j \neq \hat{h}_k$. Furthermore, if $f \in Z_{n,r}$, then by restriction we obtain an upper triangular symplectic injective partial transformation g in $\mathbb{B}_{n-1,r-1}$ in which $\hat{g}_j = f$ when $j = f(2n)$.

Thus $Z_{n,r}$ is a disjoint union of sets $\{\hat{g}_j \mid j \in E(g)\}$ as g runs over $\mathbb{B}_{n-1,r-1}$. Hence

$$|Z_{n,r}| = (2n - 2r)b_{n-1,r-1}.$$

The proof is complete. □

Lemma 3.4. Let $n \geqslant 1$, let $f \in \mathrm{RSp}_n$ and let g be its anti-transpose. Then $f \in \mathbb{B}_{n,r}$ iff $g \in \mathbb{B}_{n,r}$ for $0 \leqslant r \leqslant n$.

Proof. Write $f = [a_{i\,j}]$. Clearly, f is upper triangular iff its anti-transpose $g = [a_{\theta(j)\ \theta(i)}]$ is upper triangular. Note that $D(g) = R(f)$ and $R(g) = D(f)$. It follows that f is symplectic iff g is symplectic. The desired result follows. □

We describe the order of the upper triangular symplectic Renner monoid $\mathbb{B}_n$. Let $n \geqslant 1$ and $b_n = |\mathbb{B}_n|$, and we agree that $b_0 = 2$. Recall that $b_{n,r} = |\mathbb{B}_{n,r}|$ for $0 \leqslant r \leqslant n$. Then $b_n = 1 + \sum_{r=0}^{n} b_{n,r}$ for $n \geqslant 0$. Note that $b_{0,0} = 1$ and $b_{n,r} = 0$ if $r > n$ or $r < 0$.

Theorem 3.5. Let $n \geq 1$. Then

1) $b_{n,r} = b_{n-1,r} + (4n - 4r + 3)b_{n-1,\ r-1} + 4(n - r + 1)^2 b_{n-1,\ r-2}$, for $n \geqslant 1$ and $0 \leqslant r \leqslant n$.

2) $b_n = 4b_{n-1} + 4\big[\sum_{r=1}^{n-1}(n - r)(n - r + 1)b_{n-1,r-1}\big] - 3$, for $n \geqslant 1$.

3) $b_n = 1 + \sum_{r=0}^{n-1} 4\big[1 + (n - r)(n - r - 1)\big]b_{n-1,r}$, for $n \geqslant 1$.

Proof. We prove 1). If $n \geqslant 1$ and $r = 0$, then 1) is valid since

$$b_{n,0} = 1, \quad b_{n-1,0} = 1,\ b_{n-1,-1} = 0 = b_{n-1,-2}.$$

We now suppose that $1 \leqslant r \leqslant n$. Thus, $\mathbb{B}_{n,r} = X_{n,r} \sqcup Y_{n,r}$ where

$$X_{n,r} = \{f \in \mathbb{B}_{n,r} \mid 1 \notin R(f)\} \quad \text{and} \quad Y_{n,r} = \{f \in \mathbb{B}_{n,r} \mid 1 \in R(f)\}.$$

We first calculate $|X_{n,r}|$. If $f \in X_{n,r}$, then the first row of f is zero since $1 \notin R(f)$, and hence f takes the form

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ & * & * & \cdots & * & * \\ & & * & \cdots & * & * \\ & & & \ddots & \vdots & \vdots \\ & & & & * & * \\ & & & & & * \end{pmatrix}_{2n\times 2n} \in X_{n,r} \ ,$$

Furthermore, if $\bar{1} \in D(f)$, then there exists exactly one j such that $f(\bar{1}) = j$, where $j \neq 1$ since again $1 \notin R(f)$, so that $2 \leqslant j \leqslant 2n$. Partition $X_{n,r} = X^1_{n,r} \sqcup X^2_{n,r} \sqcup X^3_{n,r}$ where

$$\begin{aligned} X^1_{n,r} &= \{f \in \mathbb{B}_{n,r} \mid 1 \notin R(f) \text{ and } \bar{1} \notin D(f)\}. \\ X^2_{n,r} &= \{f \in \mathbb{B}_{n,r} \mid 1 \notin R(f) \text{ and } f(\bar{1}) = j \text{ for some } 2 \leqslant j \leqslant 2n-1\}. \\ X^3_{n,r} &= \{f \in \mathbb{B}_{n,r} \mid 1 \notin R(f) \text{ and } f(\bar{1}) = 2n\}. \end{aligned}$$

By Lemma 3.2, we obtain

$$|X^1_{n,r}| = b_{n-1,r}.$$

From Lemma 3.3 it follows that

$$|X^2_{n,r}| = (2n-2r)b_{n-1,r-1}.$$

If $f \in X^3_{n,r}$, then its first row is zero and $f(\bar{1}) = 2n$. Hence f is of the form

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ & * & * & \cdots & * & 0 \\ & & * & \cdots & * & 0 \\ & & & \ddots & \vdots & \vdots \\ & & & & * & 0 \\ & & & & & 1 \end{pmatrix}_{2n\times 2n} \in X^3_{n,r} \ .$$

A similar argument to that in Lemma 3.2 yields

$$|X^3_{n,r}| = b_{n-1,r-1}.$$

From $|X_{n,r}| = |X^1_{n,r}| + |X^2_{n,r}| + |X^3_{n,r}|$, it follows that

$$|X_{n,r}| = b_{n-1,r} + (2n-2r+1)b_{n-1,\ r-1}, \quad 1 \leqslant r \leqslant n. \tag{1}$$

We now calculate the cardinality of $Y_{n,r} = \{f \in \mathbb{B}_{n,r} \mid 1 \in R(f)\}$, where again $1 \leqslant r \leqslant n$. If $f \in Y_{n,r}$, then there is exactly one 1 in the first row of f. By the

admissibility of $R(f)$, the last row of f must be zero. Therefore, f is of the form

$$\begin{pmatrix} & & & \overset{\text{column } i}{\downarrow} & & & \\ 0 & 0 & \cdots & 1 & \cdots & 0 & 0 \\ & * & \cdots & 0 & \cdots & * & * \\ & & \ddots & \vdots & \cdots & \vdots & \vdots \\ & & & 0 & \cdots & * & * \\ & & & & \ddots & \vdots & \vdots \\ & & & & & * & * \\ & & & & & & 0 \end{pmatrix}_{2n\times 2n} \in Y_{n,r}\ ,$$

where $*$ is either 0 or 1, each row and each column of f contains at most one 1, and $|D(f)| = |R(f)| = r$.

If $f \in Y^4_{n,r}$ we have the following four exhaustive and mutually disjoint cases:

a) $\bar{1} \notin D(f)$ and $f(1) = 1$, where the latter implies the former by the admissibility of $D(f)$.

b) $\bar{1} \notin D(f)$ and $f(i) = 1$ for some $2 \leqslant i \leqslant 2n-1$.

c) $\bar{1} \in D(f)$ and $f(\bar{1}) = 1$.

d) $\bar{1} \in D(f)$ and $f(\bar{1}) = j$ for some $2 \leqslant j \leqslant 2n-1$.

We can divide $Y_{n,r}$ into four disjoint subsets. Let

$$\begin{aligned} Y^1_{n,r} &= \{f \in \mathbb{B}_{n,r} \mid f(1) = 1\}. \\ Y^2_{n,r} &= \{f \in \mathbb{B}_{n,r} \mid f(i) = 1 \text{ for some } 2 \leqslant i \leqslant 2n-1 \text{ and } \bar{1} \notin D(f)\}. \\ Y^3_{n,r} &= \{f \in \mathbb{B}_{n,r} \mid f(\bar{1}) = 1\}. \\ Y^4_{n,r} &= \{f \in \mathbb{B}_{n,r} \mid f(i) = 1 \text{ and } f(\bar{1}) = j \text{ for some } 2 \leqslant i,\, j \leqslant 2n-1\}. \end{aligned}$$

Then $Y_{n,r} = Y^1_{n,r} \sqcup Y^2_{n,r} \sqcup Y^3_{n,r} \sqcup Y^4_{n,r}$.

We first calculate $|Y^1_{n,r}|$. Let $f \in Y^1_{n,r}$. Then $f(1) = 1$ and $\bar{1} \notin D(f)$. Thus f takes the form

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ & * & * & \cdots & * & 0 \\ & & * & \cdots & * & 0 \\ & & & \ddots & \vdots & \vdots \\ & & & & * & 0 \\ & & & & & 0 \end{pmatrix}_{2n\times 2n} \in Y^1_{n,r}\ .$$

The matrices in $Y^1_{n,r}$ are exactly the anti-transposes of those in $X^3_{n,r}$. From Lemma 3.4 we obtain

$$|Y^1_{n,r}| = |X^3_{n,r}| = b_{n-1,r-1}, \quad 1 \leqslant r \leqslant n. \tag{2}$$

We then calculate $|Y^2_{n,r}|$. First assume that $n \geqslant 2$. Let $f \in Y^2_{n,r}$. Then $f(i) = 1$ for some $i \in \{2, \ldots, 2n-1\}$ and $f(\bar{1})$ is undefined. Therefore, the last column of f is zero,

and f is of the form

$$\begin{pmatrix} & & & \overset{\text{column } i}{\downarrow} & & & \\ 0 & 0 & \cdots & 1 & \cdots & 0 & 0 \\ & * & \cdots & 0 & \cdots & * & 0 \\ & & \ddots & \vdots & \cdots & \vdots & \vdots \\ & & & 0 & \cdots & * & 0 \\ & & & & \ddots & \vdots & \vdots \\ & & & & & * & 0 \\ & & & & & & 0 \end{pmatrix}_{2n\times 2n} \in Y^2_{n,r}\ ,$$

where $2 \leqslant i \leqslant 2n-1$. Notice that the matrices in $Y^2_{n,r}$ are exactly the anti-transposes of those in $X^2_{n,r}$. By Lemma 3.4 and Lemma 3.3 we obtain

$$|Y^2_{n,r}| = (2n-2r)b_{n-1,r-1}, \quad 1 \leqslant r \leqslant n. \tag{3}$$

When $n = 1 = r$ then $Y^2_{n,r} = \varnothing$, hence the formula is also valid when $n = 1$.

We next calculate $|Y^3_{n,r}|$. If $f \in Y^3_{n,r}$, then $f(\bar{1}) = 1$ and f is of the form

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 1 \\ & * & * & \cdots & * & 0 \\ & & * & \cdots & * & 0 \\ & & & \ddots & \vdots & \vdots \\ & & & & * & 0 \\ & & & & & 0 \end{pmatrix}_{2n\times 2n} \in Y^3_{n,r}\ .$$

A similar argument to that in Lemma 3.2 yields that the map $f \mapsto f'$ is a bijection of $Y^3_{n,r}$ onto $\mathbb{B}_{n-1,r-1}$. Therefore,

$$|Y^3_{n,r}| = b_{n-1,r-1}, \quad 1 \leqslant r \leqslant n. \tag{4}$$

We now calculate $|Y^4_{n,r}|$. Recall

$$Y^4_{n,r} = \{f \in \mathbb{B}_{n,r} \mid f(i) = 1 \text{ and } f(\bar{1}) = j \text{ for some } 2 \leqslant i, j \leqslant 2n-1\}.$$

If $f \in Y^4_{n,r}$, then the $(1,1)$, $(1,2n)$, $(2n,2n)$ entries of f are 0; hence f has the form

$$\begin{array}{c} \text{column } i \\ \downarrow \end{array}$$

$$\begin{pmatrix}
0 & 0 & \cdots & 1 & \cdots & 0 & \cdots & 0 & 0 \\
 & * & \cdots & 0 & \cdots & * & \cdots & * & 0 \\
 & & \ddots & \vdots & \cdots & \vdots & \cdots & \vdots & \vdots \\
 & & & 0 & \cdots & * & \cdots & * & 0 \\
 & & & & 0 & 0 & \cdots & 0 & 1 \\
 & & & & & * & \cdots & * & 0 \\
 & & & & & & \ddots & \vdots & \vdots \\
 & & & & & & & * & 0 \\
 & & & & & & & & 0
\end{pmatrix}_{2n\times 2n}
\begin{array}{l} \\ \\ \\ \\ \leftarrow \text{row } j \\ \\ \\ \\ \\ \end{array}
\qquad \in Y^4_{n,r} \ ,$$

where $2 \leqslant i,\, j \leqslant 2n-1$.

First, let $2 \leqslant r \leqslant n$. Deleting the first row, first column, last row, and last column of each $f \in Y^4_{n,r}$ produces an element $f' \in \mathbb{B}_{n-1,\,r-2}$. We have a map $\kappa\colon Y^4_{n,r} \to \mathbb{B}_{n-1,\,r-2}$. We first show that for each $g \in \mathbb{B}_{n-1,\,r-2}$, there are $2(n-r+1)$ distinct f's in $Y^4_{n,r}$ such that $f' = g$. Indeed fix $g \in \mathbb{B}_{n-1,\,r-2}$. Let $I = \tau^{-1}(D(g))$ and $J = \tau^{-1}(R(g))$, where $\tau(k) = k-1$. By Lemma 2.4 a), I and J are admissible in $\mathcal{A}_n$, and $|I \cup \theta(I)| = |J \cup \theta(J)| = 2r-4$, where θ is the involution in $\mathbf{2n}$. Let $E(g)$ and $F(g)$ denote the complements of $I \cup \theta(I)$ and $J \cup \theta(J)$ in $\mathbf{2n} \setminus \{1, 2n\}$, respectively, so that $|E(g)| = |F(g)| = 2(n-r+1) \geqslant 2$.

For each $i \in E(g)$ and $j \in F(g)$ we extend g by adding a zero row at the bottom, a zero column at the left, and a row 1 with 1 in position i, and a column $2n$ with 1 in position j; this extension is denoted by $\hat{g}_{ij}$. The domain $D(\hat{g}_{ij}) = \tau^{-1}(D(g)) \cup \{i\}$ and the range $R(\hat{g}_{ij}) = \tau^{-1}(R(g)) \cup \{j\}$ are also admissible in $\mathcal{A}_n$ since both $\theta(D(\hat{g}_{ij})) \cap D(\hat{g}_{ij})$ and $\theta(R(\hat{g}_{ij})) \cap R(\hat{g}_{ij})$ are empty by the construction of $E(f)$, $E(g)$ and that $\theta(i) \neq i$, $\theta(j) \neq j$. Thus $\hat{g}_{ij}$ is in $Y^4_{n,r}$. Moreover $\kappa(\hat{g}_{ij}) = g$ for all $(i,j) \in E(g) \times F(g)$.

Note that if $(i,j) \neq (i_1, j_1)$ in $E(g) \times F(g)$, then $\hat{g}_{ij} \neq \hat{g}_{i_1 j_1}$. Moreover, if $g \neq h$ in $\mathbb{B}_{n-1,\,r-2}$ then for each pair $(i,j) \in E(g) \times F(g)$ and $(k,l) \in E(h) \times F(h)$ we have $\hat{g}_{ij} \neq \hat{h}_{kl}$. Furthermore, if $f \in Y^4_{n,r}$ then by restriction we obtain $g = f' \in \mathbb{B}_{n-1,\,r-2}$ in which $f = \hat{g}_{ij}$ where $f(i) = 1$ and $j = f(2n)$.

Thus $Y^4_{n,r}$ is a disjoint union of sets $\{\hat{g}_{ij} \mid i \in E(g), j \in F(g)\}$ as g runs over $\mathbb{B}_{n-1,\,r-2}$. We obtain, for $2 \leqslant r \leqslant n$

$$|Y^4_{n,r}| = 4(n-r+1)^2\, b_{n-1,\,r-2}. \tag{5}$$

Now, let $n \geqslant 1$ and $r = 1$. Then $Y^4_{n,r} = Y^4_{n,1} = \varnothing$; hence $|Y^4_{n,1}| = 0$. Note that in this case $b_{n-1,\,r-2} = b_{n-1,\,-1} = 0$. Thus formula (5) still holds.

Since $|Y_{n,r}| = |Y^1_{n,r}| + |Y^2_{n,r}| + |Y^3_{n,r}| + |Y^4_{n,r}|$, by (2), (3), (4), and (5) we obtain

$$|Y_{n,r}| = 2(n-r+1)b_{n-1,\ r-1} + 4(n-r+1)^2 b_{n-1,\ r-2}, \quad 1 \leqslant r \leqslant n. \tag{6}$$

It follows from (1) and (6) that

$$\begin{aligned} b_{n,\,r} &= |X_{n,\,r}| + |Y_{n,\,r}| \\ &= b_{n-1,\,r} + (4n-4r+3)b_{n-1,\ r-1} + 4(n-r+1)^2 b_{n-1,\ r-2}, \quad 1 \leqslant r \leqslant n. \end{aligned} \tag{7}$$

Moreover when $n \geqslant 1$ and $r = 0$ equality (7) still holds since $b_{0,\,0} = 1$ and $b_{n,\,r} = 0$ if $n < r$ or $r < 0$, so that both sides are equal to 1. The proof of 1) is complete.

We can now prove 2). Recall that $\mathbb{B}_n = \big(\bigsqcup_{r=0}^{n} \mathbb{B}_{n,\,r}\big) \bigsqcup \{\mathrm{id}_{2n}\}$. From (7) we obtain

$$\begin{aligned} b_n &= 1 + \sum_{r=0}^{n-1} b_{n-1,\,r} + 3\sum_{r=1}^{n} b_{n-1,\,r-1} + 4\sum_{r=1}^{n}(n-r)b_{n-1,\ r-1} + 4\sum_{r=2}^{n}(n-r+1)^2 b_{n-1,\,r-2} \\ &= 4\Big(1 + \sum_{r=0}^{n-1} b_{n-1,\,r}\Big) + 4\sum_{r=1}^{n}(n-r)b_{n-1,\,r-1} + 4\Big[\sum_{r=2}^{n}(n-r+1)^2 b_{n-1,\,r-2}\Big] - 3 \\ &= 4b_{n-1} + 4\sum_{r=1}^{n-1}(n-r)b_{n-1,\ r-1} + 4\Big[\sum_{r=1}^{n-1}(n-r)^2 b_{n-1,\,r-1}\Big] - 3 \\ &= 4b_{n-1} + 4\Big[\sum_{r=1}^{n-1}(n-r)(n-r+1)b_{n-1,\,r-1}\Big] - 3\ , \quad \text{for } n \geqslant 1. \end{aligned}$$

The proof of 2) is complete.

We now prove 3). From 2) it follows that

$$\begin{aligned} b_n &= 4b_{n-1} + \Big[\sum_{r=1}^{n-1} 4(n-r+1)(n-r)b_{n-1,\,r-1}\Big] - 3 \\ &= 4\Big(1 + \sum_{r=0}^{n-1} 4b_{n-1,\,r}\Big) + \Big[\sum_{r=0}^{n-2} 4(n-r)(n-r-1)b_{n-1,\,r}\Big] - 3 \\ &= 1 + 4b_{n-1,\,n-1} + \sum_{r=0}^{n-2} 4b_{n-1,\,r} + \sum_{r=0}^{n-2} 4(n-r)(n-r-1)b_{n-1,\,r} \\ &= 1 + 4b_{n-1,\,n-1} + \sum_{r=0}^{n-2} 4\big[1 + (n-r)(n-r-1)\big]b_{n-1,\,r} \\ &= 1 + \sum_{r=0}^{n-1} 4\big[1 + (n-r)(n-r-1)\big]b_{n-1,\,r} \end{aligned}$$

where the last equality follows from $1 + (n-r)(n-r-1) = 1$ when $r = n-1$. □

The first seven cases of $b_{n,\,r}$ and b_n are provided in the following table.

4 Generating Functions and Closed Formulas

4.1 The Exponential Generating Function of b_n

We investigate the exponential generating function of b_n. Recall that $b_n = |\mathbb{B}_n|$ for $n \geqslant 1$, and $b_0 = 2$.

$b_{n,r}$	r: 0	1	2	3	4	5	6	7	2n	b_n
n:1	1	3	1							5
2	1	10	13		1					25
3	1	21	99	79			1			201
4	1	36	366	1108	633				1	2145
5	1	55	970	6430	14245	6331			1	28033
6	1	78	2115	24500	119895	208926	75973		1	431489
7	1	105	4053	72485	622755	2409771	3456775	1063623	1	7629569

Theorem 4.1. The series $\sum_{n=0}^{\infty}(b_n/n!)t^n$ converges for $|t| < 1/2$ to the function

$$b(t) = e^t\Big[1 + \frac{e^{t/(1-2t)}}{1-2t}\Big].$$

Proof. We first investigate the exponential generating function of $s_n := b_n - 1$, where $n \geq 0$. Then,

$$s_n = \sum_{r=0}^{n} b_{n,\,r}, \quad \text{with} \quad s_0 = 1.$$

By Theorem 3.5 part 1), if $1 \leqslant n$ and $0 \leqslant r \leqslant n$ then

$$b_{n,\,r} = b_{n-1,\,r} + (4n - 4r + 3)b_{n-1,\,r-1} + 4(n - r + 1)^2 b_{n-1,\,r-2}\ ,$$

where $b_{0,\,0} = 1$ and $b_{j,\,k} = 0$ for $j < k$ or $k < 0$.

Let $s(t) = \sum_{n=0}^{\infty}(s_n/n!)t^n$. We claim that $s(t)$ converges for $|t| < 1/2$ to the function

$$\frac{e^t e^{t/(1-2t)}}{1-2t}.$$

Indeed, write $s_{n,\,k} = b_{n,\,n-k}$ for $0 \leqslant k \leqslant n$. Then $b_{n,\,r} = s_{n,\,n-r}$, $s_{0,\,0} = 1$, and $s_{j,\,k} = 0$ for $j < k$ or $k < 0$. Thus

$$s_{n,\,k} = s_{n-1,\,k-1} + (4k + 3)s_{n-1,\,k} + 4(k + 1)^2 s_{n-1,\,k+1}, \quad 1 \leqslant n; 0 \leqslant k \leqslant n. \tag{8}$$

Multiplying by x^k and summing, we obtain

$$\sum_{k=0}^{n} s_{n,\,k}\ x^k = \sum_{k=0}^{n} s_{n-1,\,k-1}\ x^k + \sum_{k=0}^{n}(4k + 3)s_{n-1,\,k}\ x^k + \sum_{k=0}^{n} 4(k + 1)^2 s_{n-1,\,k+1}\ x^k. \tag{9}$$

Define polynomials

$$S_n(x) = \sum_{k=0}^{n} s_{n,\,k}\ x^k, \quad n \geqslant 0. \tag{10}$$

Then

$$S_{n-1}(x) = \sum_{k=0}^{n-1} s_{n-1,\,k}\ x^k;\ S'_{n-1}(x) = \sum_{k=1}^{n-1} k s_{n-1,\,k}\ x^{k-1};\ S''_{n-1}(x) = \sum_{k=2}^{n-1} k(k-1) s_{n-1,\,k}\ x^{k-2}.$$

The three sums on the right of (9) are

$$\sum_{k=0}^{n} s_{n-1,\,k-1}\, x^k = x \sum_{k=-1}^{n-1} s_{n-1,\,k}\, x^k = xS_{n-1}(x)\,,$$

$$\sum_{k=0}^{n} (4k+3) s_{n-1,\,k}\, x^k = \sum_{k=0}^{n-1} (4k+3) s_{n-1,\,k}\, x^k = 4xS'_{n-1}(x) + 3S_{n-1}(x)\,,$$

$$\begin{aligned}\sum_{k=0}^{n} 4(k+1)^2 s_{n-1,\,k+1}\, x^k &= \sum_{k=1}^{n+1} 4k^2 s_{n-1,\,k}\, x^{k-1} \\ &= 4x \sum_{k=1}^{n-1} k(k-1) s_{n-1,\,k}\, x^{k-2} + 4 \sum_{k=1}^{n-1} k s_{n-1,\,k}\, x^{k-1} \\ &= 4xS''_{n-1}(x) + 4S'_{n-1}(x)\,.\end{aligned}$$

Therefore, (9) becomes

$$S_n(x) = 4xS''_{n-1}(x) + (4x+4)S'_{n-1}(x) + (x+3)S_{n-1}(x), \quad n \geqslant 1.$$

We have

$$\sum_{n=1}^{\infty} \frac{S_n(x)t^{n-1}}{(n-1)!} = 4x \sum_{n=1}^{\infty} \frac{S''_{n-1}(x)t^{n-1}}{(n-1)!} + (4x+4) \sum_{n=1}^{\infty} \frac{S'_{n-1}(x)t^{n-1}}{(n-1)!} + (x+3) \sum_{n=1}^{\infty} \frac{S_{n-1}(x)t^{n-1}}{(n-1)!}.$$

Define the function

$$u(x,\,t) = \sum_{n=0}^{\infty} \frac{S_n(x)t^n}{n!} \tag{11}$$

so that $u(x,\,0) = S_0(x) = s_{0,\,0} = 1$. We obtain a PDE with an auxiliary condition:

$$\frac{\partial u}{\partial t} = 4x\frac{\partial u^2}{\partial x^2} + (4x+4)\frac{\partial u}{\partial x} + (x+3)u, \quad u(x,\,0) = 1.$$

We seek a solution in the form $u(x,\,t) = \beta(t)e^{x\alpha(t)}$. The auxiliary condition gives

$$1 = \beta(0)e^{x\alpha(0)} \quad \text{for all } x.$$

Hence $\alpha(0) = 0$ and $\beta(0) = 1$. Substituting the ansatz in the differential equation, we find

$$e^{x\alpha}(\beta' + x\beta\alpha') = 4x\alpha^2\beta e^{x\alpha} + (4x+4)\alpha\beta e^{x\alpha} + (x+3)\beta e^{x\alpha}.$$

Equating coefficients of $xe^{x\alpha}$ and $e^{x\alpha}$ from both sides, we obtain two equations, namely

$$\alpha' = (2\alpha+1)^2, \quad \alpha(0) = 0\,;$$

$$\beta' = (4\alpha+3)\beta, \quad \beta(0) = 1\,.$$

The solutions are

$$\alpha = \frac{t}{1-2t} \quad \text{and} \quad \beta = \frac{e^t}{1-2t}, \quad t < \tfrac{1}{2}.$$

Hence

$$u(x,\, t) = \frac{e^t e^{tx/(1-2t)}}{1-2t}, \quad |t| < \tfrac{1}{2},\ x \in \mathrm{R}\,. \tag{12}$$

Note that from (10) we have

$$S_n(1) = \sum_{k=0}^{n} s_{n,\,k} = \sum_{r=0}^{n} b_{n,\,r} = s_n, \quad n \geqslant 0.$$

From (11) and (12) it follows that

$$s(t) = \sum_{n=0}^{\infty} \frac{s_n\, t^n}{n!} = u(1,\, t) = \frac{e^t e^{t/(1-2t)}}{1-2t}, \tag{13}$$

and the series is convergent for $|t| < \frac{1}{2}$. The claim holds.

Next, note that $b_n = 1 + s_n$. We obtain

$$\begin{aligned} b(t) &= \sum_{n=0}^{\infty} b_n \frac{t^n}{n!} = \sum_{n=0}^{\infty} \frac{t^n}{n!} + \sum_{n=0}^{\infty} s_n \frac{t^n}{n!} = e^t + s(t) \\ &= e^t + \frac{e^t e^{t/(1-2t)}}{1-2t}\ . \end{aligned}$$

The desired result follows. □

4.2 An Explicit Closed Formula for b_n

Theorem 4.2. Let $b_n = |\mathbb{B}_n|$ for $n \geqslant 1$. Then

$$b_n = 1 + \sum_{k=0}^{n} \left\{ \sum_{l=0}^{k} \binom{n}{k} \binom{k}{l} \frac{2^k\, k!}{2^l\ l!} \right\}, \quad n \geqslant 1\,.$$

Proof. The Laguerre polynomials are

$$L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n}(x^n e^{-x}) = \sum_{k=0}^{n} \frac{n!(-x)^k}{(k!)^2(n-k)!}\,, \quad x \in \mathrm{R},\ n \geqslant 0.$$

Their exponential generating function is

$$\sum_{n=0}^{\infty} L_n(x) z^n = \frac{e^{-xz/(1-z)}}{1-z}, \quad x \in \mathrm{R},\ |z| < 1.$$

Substituting $x = -\frac{1}{2}$ and $z = 2t$, we obtain

$$\sum_{n=0}^{\infty} L_n(-\tfrac{1}{2})(2t)^n = \frac{e^{t/(1-2t)}}{1-2t}, \quad |t| < \tfrac{1}{2}.$$

Substituting this in (13), we have

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{s_n t^n}{n!} = e^t\sum_{n=0}^{\infty}L_n(-\tfrac{1}{2})2^n t^n &= \Big(\sum_{n=0}^{\infty}\frac{t^n}{n!}\Big)\Big(\sum_{n=0}^{\infty}L_n(-\tfrac{1}{2})2^n t^n\Big)\\ &= \sum_{n=0}^{\infty}\Big\{\sum_{k=0}^{n}\frac{L_k(-\frac{1}{2})2^k}{(n-k)!}\Big\}t^n \qquad (14)\\ &= \sum_{n=0}^{\infty}\Big\{\sum_{k=0}^{n}\sum_{l=0}^{k}\frac{k!2^{k-l}}{(l!)^2(n-k)!(k-l)!}\Big\}t^n.\end{aligned}$$

Hence

$$s_n = \sum_{k=0}^{n}\sum_{l=0}^{k}\frac{n!k!2^{k-l}}{(l!)^2(n-k)!(k-l)!} = \sum_{k=0}^{n}\Big\{\sum_{l=0}^{k}\binom{n}{k}\binom{k}{l}\frac{2^k k!}{2^l l!}\Big\}.$$

As $b_n = 1 + s_n$ we obtain our result. □

The order b_n of the upper triangular symplectic Renner monoids is closely related to the Laguerre polynomials.

Corollary 4.3. Let $L_k(t)$ be the kth Laguerre polynomial. Then

$$b_n = 1 + n!\sum_{k=0}^{n}\frac{2^k}{(n-k)!}L_k\left(-\tfrac{1}{2}\right), \quad n \geqslant 0.$$

Proof. The proof follows from comparing the coefficients of t^n on both sides of (14). □

We shall obtain a recurrence relation for the sequence b_n involving its immediate past three terms: b_{n-1}, b_{n-2}, and b_{n-3}.

Theorem 4.4. Let $b_n = 1 + \sum_{r=0}^{n} b_{n,r}$, where $n \geqslant 0$. Then $b_0 = 2$, $b_1 = 5$, $b_2 = 25$ and

$$b_n = 4nb_{n-1} - 4n(n-1)b_{n-2} + 4(n-1)(n-2)b_{n-3} + 4n - 7, \quad n \geqslant 3.$$

Proof. Let $s_n = \sum_{r=0}^{n} b_{n,r}$. From (13) it follows that

$$s(t) := \sum_{n=0}^{\infty}\frac{s_n\, t^n}{n!} = \frac{e^t e^{t/(1-2t)}}{1-2t}, \quad |t| < \tfrac{1}{2}.$$

A straightforward calculation shows that

$$\frac{s'(t)}{s(t)} = \frac{4(1-t)^2}{(1-2t)^2}.$$

That is,

$$(4t^2 - 4t + 1)s'(t) = (4t^2 - 8t + 4)s(t).$$

The left side is

$$4\sum_{n=2}^{\infty}\frac{n(n-1)s_{n-1}t^n}{n!} - 4\sum_{n=1}^{\infty}\frac{ns_n t^n}{n!} + \sum_{n=1}^{\infty}\frac{s_{n+1}t^n}{n!}.$$

The right side is

$$\begin{aligned}
&4\sum_{n=0}^{\infty}\frac{s_n t^{n+2}}{n!}-8\sum_{n=0}^{\infty}\frac{s_n t^{n+1}}{n!}+4\sum_{n=0}^{\infty}\frac{s_n t^n}{n!}\\
&\quad=4\sum_{n=2}^{\infty}\frac{s_{n-2}t^n}{(n-2)!}-8\sum_{n=1}^{\infty}\frac{s_{n-1}t^n}{(n-1)!}+4\sum_{n=0}^{\infty}\frac{s_n t^n}{n!}\\
&\quad=4\sum_{n=2}^{\infty}\frac{n(n-1)s_{n-2}t^n}{n!}-8\sum_{n=1}^{\infty}\frac{ns_{n-1}t^n}{n!}+4\sum_{n=0}^{\infty}\frac{s_n t^n}{n!}.
\end{aligned}$$

Comparing coefficients of t^0, t gives $s_1 = 4s_0 = 4$ and $-4s_1 + s_2 = -8s_0 + 4s_1$ so $s_2 = 24$. Whereas, for $n \geqslant 2$ we have

$$4n(n-1)s_{n-1} - 4ns_n + s_{n+1} = 4n(n-1)s_{n-2} - 8ns_{n-1} + 4s_n$$

which rearranges to

$$s_n = 4ns_{n-1} - 4n(n-1)s_{n-2} + 4(n-1)(n-2)s_{n-3}, \quad n \geqslant 3.$$

Since $b_n = s_n + 1$, we obtain $b_0 = 2,\ b_1 = 5,\ b_2 = 25$ and

$$b_n = 4nb_{n-1} - 4n(n-1)b_{n-2} + 4(n-1)(n-2)b_{n-3} + 4n - 7, \quad n \geqslant 3. \qquad \square$$

4.3 The Exponential Generating Function of $b_{n,\,n-r}$

We now fix an $r \geqslant 0$, and consider the exponential generating function for the sequence $n \mapsto b_{n,n-r}$. We note that $\sum\limits_{n=0}^{\infty}\frac{b_{n,n-r}\ t^n}{n!} = \sum\limits_{n=r}^{\infty}\frac{b_{n,n-r}\ t^n}{n!}$.

Theorem 4.5. Let $n \geqslant 1$ and let $b_{n,r} = |\mathbb{B}_{n,r}|$ for $0 \leqslant r \leqslant n$. Then

$$\sum_{n=r}^{\infty}\frac{b_{n,n-r}\ t^n}{n!} = \frac{t^r e^t}{r!(1-2t)^{r+1}}, \qquad |t| < \tfrac{1}{2},\ r \geqslant 0.$$

Proof. Fix $r \geqslant 0$. Write $s_{n,\,r} = b_{n,\,n-r}$ so that

$$\sum_{n=r}^{\infty}\frac{s_{n,\,r}\ t^n}{n!} = \sum_{n=r}^{\infty}\frac{b_{n,n-r}\ t^n}{n!}. \tag{15}$$

By (10), (11) and (12), and the series expansion for $\exp x$ we have

$$\sum_{n=0}^{\infty}\Big\{\sum_{r=0}^{n}\frac{s_{n,r}\,x^r t^n}{n!}\Big\} = \frac{e^t e^{tx/(1-2t)}}{1-2t} = \frac{e^t}{1-2t}\sum_{r=0}^{\infty}\frac{t^r x^r}{r!(1-2t)^r}. \tag{16}$$

An interchange of order of summation and (15) gives

$$\sum_{n=0}^{\infty}\Big\{\sum_{r=0}^{n}\frac{s_{n,r}\,x^r t^n}{n!}\Big\} = \sum_{r=0}^{\infty}\Big\{\sum_{n=r}^{\infty}\frac{s_{n,r}\ x^r t^n}{n!}\Big\} = \sum_{r=0}^{\infty}\sum_{n=r}^{\infty}\frac{b_{n,n-r}\ x^r t^n}{n!}.$$

Hence

$$\sum_{r=0}^{\infty}\Big\{\sum_{n=r}^{\infty}\frac{b_{n,n-r}\,t^n}{n!}\Big\}x^r=\frac{e^t}{1-2t}\sum_{r=0}^{\infty}\frac{t^r}{r!(1-2t)^r}\,x^r.$$

Comparing coefficients of x^r, we obtain

$$\sum_{n=r}^{\infty}\frac{b_{n,n-r}\,t^n}{n!}=\frac{t^re^t}{r!(1-2t)^{r+1}},\quad |t|<\tfrac{1}{2},\ r\geq 0. \tag{17}$$

The proof is complete. □

4.4 An Explicit Closed Form for Every $b_{n,r}$

Theorem 4.6. Let $n\geqslant 1$ and let $0\leqslant r\leqslant n$. Then

$$b_{n,r}=\sum_{k=0}^{r}\frac{n!\,2^k}{(n-r)!(r-k)!}\binom{n-r+k}{k}.$$

Proof. Fix $r\geqslant 0$. By (17)

$$\sum_{n=r}^{\infty}\frac{b_{n,n-r}\,t^n}{n!}=\frac{t^re^t}{r!(1-2t)^{r+1}}.$$

Fix N, R with $N\geqslant 1$ and $0\leqslant R\leqslant N$. Put $r=N-R$; we obtain

$$\sum_{n=N-R}^{\infty}\frac{b_{n,n-N+R}\,t^n}{n!}=\frac{t^Nt^{-R}e^t}{(N-R)!(1-2t)^{N-R+1}}. \tag{18}$$

In the left side the coefficient of t^N is

$$\frac{b_{N,R}}{N!}.$$

Put $\alpha=N-R+1$, so α is a positive integer. We have expansions

$$e^t=\sum_{n=0}^{\infty}\frac{t^n}{n!},\quad\text{and}\quad (1-2t)^{-\alpha}=\sum_{n=0}^{\infty}\binom{-\alpha}{n}(-2t)^n.$$

Hence

$$e^t(1-2t)^{-\alpha}=\sum_{n=0}^{\infty}c_nt^n\quad\text{where}\quad c_n=\sum_{k=0}^{n}\frac{1}{(n-k)!}\binom{-\alpha}{k}(-1)^k2^k.$$

It can easily be shown that

$$\binom{-\alpha}{k}(-1)^k=\binom{N-R+k}{k}.$$

The right side of (18) is

$$\frac{t^Nt^{-R}}{(N-R)!}\{c_0+c_1t+\cdots+c_Rt^R+\dots\}.$$

Hence

$$\frac{b_{N,R}}{N!}=\frac{c_R}{(N-R)!}=\frac{1}{(N-R)!}\sum_{k=0}^{R}\frac{2^k}{(R-k)!}\binom{N-R+k}{k}.$$

□

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